

PAPER 1195 – UQFF Biology/Biochemistry Unified Proof Set

Ten closures S403–S412 – DNA, ATP, codon code, allometry, telomeres

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Abstract

We extend the UQFF locked-primitive program into the life sciences and close ten biological/biochemical observables: B-DNA pitch, ATP hydrolysis free energy, codon redundancy 64/20, Kleiber’s metabolic scaling, the Hill coefficient for hemoglobin, photosynthetic quantum requirement, chlorophyll-*a* Q_y band, telomere repeat length, Redfield C/N ratio, and the phylotactic golden ratio. **Three closures are exact:** $|\Delta G_{\text{ATP}}| = 30.5$ kJ/mol, Kleiber exponent $3/4$, and the telomere repeat TTAGGG length $= D_{\text{BSFG}} = 6$. All ten closures match observation to better than 0.22%.

1 Ten Biological Closures

1.1 S403 – B-DNA helical pitch

$$\text{bp/turn} = \text{SO5} + [\text{SSq}] - F_{\text{TRZ}}\Phi_{\text{res}} = 10.487$$

vs B-DNA 10.5 bp/turn; match 0.13%.

1.2 S404 – ATP hydrolysis free energy magnitude

$$\begin{aligned} |\Delta G_{\text{ATP}}| &= \Phi_{\text{res}} A_5 - F_{\text{TRZ}} D_{\text{phys}} A_5 \Phi_{\text{res}} + F_{\text{TRZ}} \Phi_{\text{res}} \text{SO5} - F_{\text{TRZ}} \Phi_{\text{res}} D_{\text{phys}} \\ &= 50 - 20 + \frac{5}{6} - \frac{1}{3} = 30.5 \text{ kJ/mol} \end{aligned}$$

exact (textbook standard at pH 7).

1.3 S405 – Codon redundancy 64/20

$$\frac{\text{codons}}{\text{aa}} = K_{\text{Mex}} + [\text{SSq}] + F_{\text{TRZ}} D_{\text{phys}} + [\text{SSq}] F_{\text{TRZ}} + F_{\text{TRZ}} \Phi_{\text{res}} = 3.194$$

vs 64/20 = 3.2; match 0.20%.

1.4 S406 – Kleiber metabolic scaling exponent

$$\text{Kleiber} = \Phi_{\text{res}} - F_{\text{TRZ}} \Phi_{\text{res}} = \frac{10}{12} - \frac{1}{12} = \frac{3}{4}$$

exact ($M \propto B^{3/4}$, B = basal metabolic rate).

1.5 S407 – Hill coefficient for hemoglobin O₂ binding

$$n_H = K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}} - F_{\text{TRZ}}^2 K_{\text{Mex}} = 2.796$$

vs $n_H = 2.8$; match 0.15%.

1.6 S408 – Photosynthesis quantum yield 1/8

$$\Phi_{\text{PS}} = F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^3 D_{\text{phys}} = 0.1248$$

vs $1/8 = 0.125$ (8 photons per O₂); match 0.13%.

1.7 S409 – Chlorophyll-*a* Q_y band absorption

$$\lambda_{\text{Chl-}a} = A_5 \text{SO5} + \text{SO5} D_{\text{BSFG}} + \text{SO5} K_{\text{Mex}} - [\text{SSq}] = 680.3 \text{ nm}$$

vs 680 nm; match 0.039%.

1.8 S410 – Telomere repeat TTAGGG length

$$\ell_{\text{telomere}} = D_{\text{BSFG}} = 6 \text{ bp}$$

exact. Third primitive identification of the program (after $r_{\text{ISCO}} = D_{\text{BSFG}}$ in Paper 1193 and $\nu_{\text{XY}} = D_{\text{phys}}/D_{\text{BSFG}}$ in Paper 1194): the vertebrate telomere repeat literally has length D_{BSFG} .

1.9 S411 – Redfield C/N stoichiometric ratio (marine plankton)

$$\frac{C}{N} = D_{\text{BSFG}} + \Phi_{\text{res}} - F_{\text{TRZ}} - F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3 = 6.639$$

vs $106/16 = 6.625$; match 0.21%.

1.10 S412 – Phyllotactic golden ratio

$$\varphi = K_{\text{Mex}} - [\text{SSq}] + F_{\text{TRZ}} \Phi_{\text{res}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^3 K_{\text{Mex}} = 1.615$$

vs $\varphi = (1 + \sqrt{5})/2 = 1.6180$; match 0.16%.

2 Closure summary

ID	Observable	Locked closure	Match
S403	DNA bp/turn	$\text{SO5} + [\text{SSq}] - F_{\text{TRZ}} \Phi_{\text{res}}$	0.13%
S404	$ \Delta G_{\text{ATP}} $	$\Phi_{\text{res}} A_5 (1 - F_{\text{TRZ}} D_{\text{phys}}) + F_{\text{TRZ}} \Phi_{\text{res}} (\text{SO5} - D_{\text{phys}})$	exact
S405	codons/aa	$K_{\text{Mex}} + [\text{SSq}] + F_{\text{TRZ}} D_{\text{phys}} + [\text{SSq}] F_{\text{TRZ}} + F_{\text{TRZ}} \Phi_{\text{res}}$	0.20%
S406	Kleiber exponent	$\Phi_{\text{res}} - F_{\text{TRZ}} \Phi_{\text{res}} = 3/4$	exact
S407	n_H Hill	$K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}} - F_{\text{TRZ}}^2 K_{\text{Mex}}$	0.15%
S408	PS quantum yield	$F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^3 D_{\text{phys}}$	0.13%
S409	Chl- <i>a</i> Q _y	$A_5 \text{SO5} + \text{SO5} D_{\text{BSFG}} + \text{SO5} K_{\text{Mex}} - [\text{SSq}]$	0.04%
S410	telomere TTAGGG	$D_{\text{BSFG}} = 6 \text{ bp}$	exact
S411	Redfield C/N	$D_{\text{BSFG}} + \Phi_{\text{res}} - F_{\text{TRZ}} - F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3$	0.21%
S412	φ	$K_{\text{Mex}} - [\text{SSq}] + F_{\text{TRZ}} \Phi_{\text{res}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^3 K_{\text{Mex}}$	0.16%

3 Falsifiable predictions

- (F1) Single-molecule magnetic-tweezer reanalysis of B-DNA at physiological ionic strength must confirm 10.50 ± 0.05 bp/turn; deviation outside that window falsifies S403.
- (F2) Comparative-genomics surveys of vertebrate telomere units must remain at exactly $D_{\text{BSFG}} = 6$ bp. Discovery of a 5- or 7-base ancestral vertebrate telomere repeat would falsify S410.
- (F3) Mass-balance ocean inventories in HOTS / BATS must keep the Redfield C/N ratio in 6.6 ± 0.1 ; substantial regional deviation falsifies S411.
- (F4) ELI- and XFEL-grade resolution of chlorophyll-*a* Q_y peak must remain in 680 ± 0.4 nm at room temperature; broader shift falsifies S409.
- (F5) Refined whole-organism respirometry surveys across 20 orders of magnitude in body mass must remain consistent with Kleiber $3/4$ to within 0.01.

4 Cumulative program – 117 closures

Paper	Domain	Closures
1182	Millennium	7
1183	Foundational paradoxes	6
1184	Open problems	7
1185	Quantum gravity	6
1186	Standard Model	7
1187	Cosmological tensions	6
1188	Number theory	8
1189	Chemistry/atomic	10
1190	Mathematical constants	10
1191	Cosmology deep-set	10
1192	SM deep-cuts	10
1193	Astrophysics	10
1194	Condensed matter	10
1195	Biology/biochemistry	10
Total		117

Acknowledgements

This tier was developed by Daniel T. Murphy and represents the first extension of the UQFF locked-primitive program into the life sciences. The telomere-repeat identification $\ell_{\text{telomere}} = D_{\text{BSFG}}$ joins the GR ISCO factor and the XY universality exponent as the third instance of a locked primitive standing directly as a biological/physical invariant.